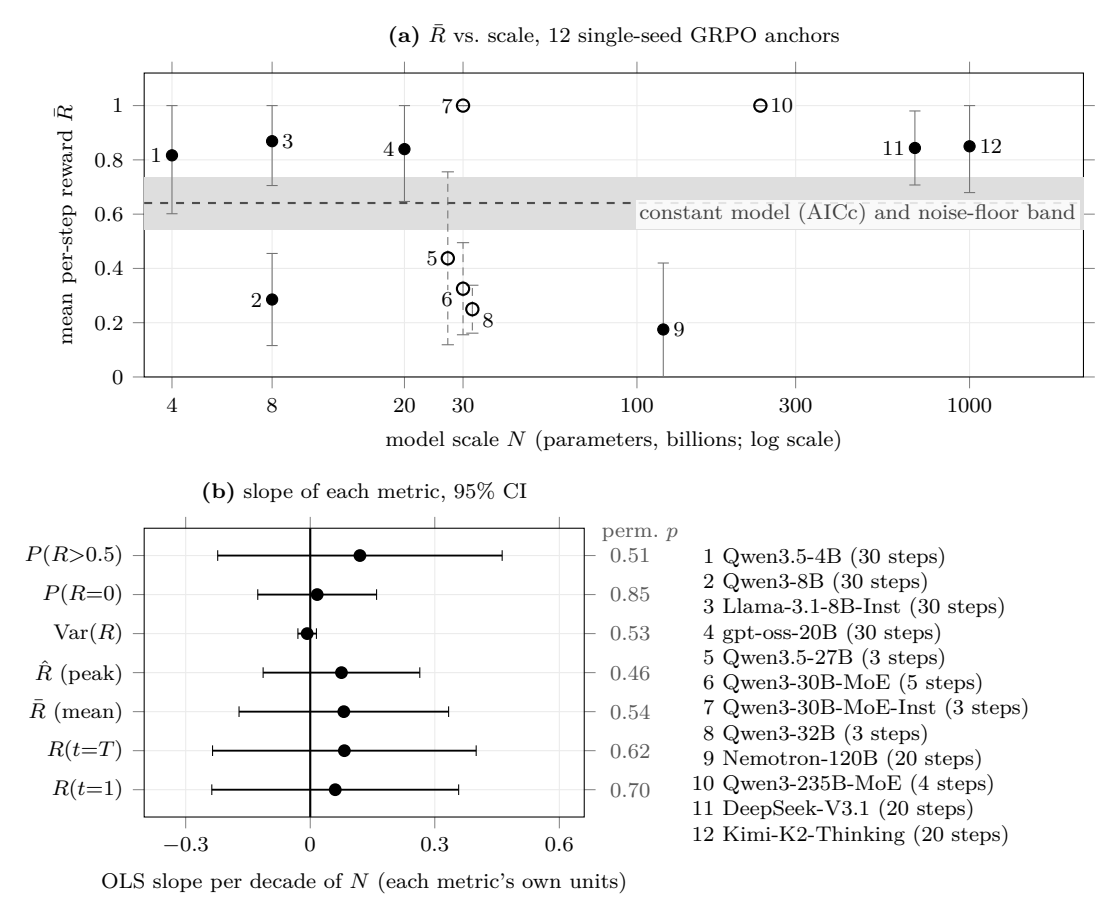
(a) $\bar{R}$ vs. scale, 12 single-seed GRPO anchors
mean per-step reward $\bar{R}$
constant model (AICc) and noise-floor band
model scale N (parameters, billions; log scale)
(b) slope of each metric, 95% CI
perm. p
$P(R>0.5)$
$P(R=0)$
Var(R)
$\hat{R}$ (peak)
$\bar{R}$ (mean)
$R(t=T)$
$R(t=1)$
0.51
0.85
0.53
0.46
0.54
0.62
0.70
OLS slope per decade of N (each metric's own units)
1 Qwen3.5-4B (30 steps)
2 Qwen3-8B (30 steps)
3 Llama-3.1-8B-Inst (30 steps)
4 gpt-oss-20B (30 steps)
5 Qwen3.5-27B (3 steps)
6 Qwen3-30B-MoE (5 steps)
7 Qwen3-30B-MoE-Inst (3 steps)
8 Qwen3-32B (3 steps)
9 Nemotron-120B (20 steps)
10 Qwen3-235B-MoE (4 steps)
11 DeepSeek-V3.1 (20 steps)
12 Kimi-K2-Thinking (20 steps)